STA 2002 Midterm Exam

October 31, 2018

19:00-21:00

1 Short Questions

(7×4 points)

1. Fei Haohan and Ying Xiong are both interested in estimating the mean of a certain normally distributed population. Being a little lazy, Fei Haohan decides that he will only a take a sample of size 4. Ying Xiong is quite diligent and takes a sample of size 100. They both compute 95% confidence intervals.

   (a) Assuming the population variance is known to be σ^2, how much wider will Fei Haohan's confidence interval be than Ying Xiong's? (Give the ratio of the width of the two intervals.)
   Solution: $\sqrt{100}/\sqrt{4} = 5$

   (b) True or false: Ying Xiong has a greater probability of capturing the true mean in her confidence interval than does Fei Haohan.
   Solution: False

2. Suppose a random sample of $n = 30$ observations is drawn from a normally distributed population and the sample mean and standard deviation are $\bar{x} = 10$ and $s^2 = 17.04$.

   What is the p-value for a hypothesis test of $H_0 : \sigma^2 = 25$ versus $H_1 : \sigma^2 > 25$?

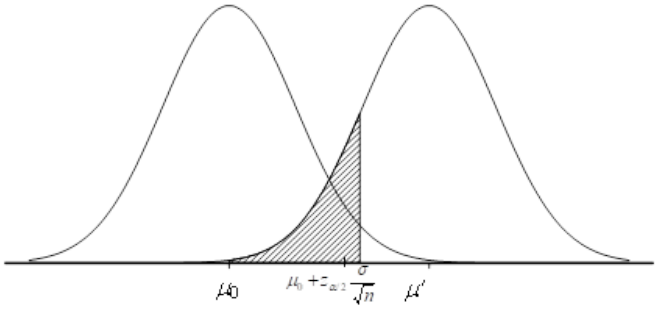
   (a) 0.1

   (b) 0.9

   (c) 0.8

   (d) 0.05

   Solution: (b). The test statistic is $\chi^2 = \frac{(n-1)S^2}{\sigma^2}$ and has a χ^2-distribution with $n - 1 = 29$ degrees of freedom. $P(\chi^2 > \frac{29 \cdot 17.04}{25}) = P(\chi^2 > 19.77) = 0.9$.



3. In a hypothesis test of $H_0 : \mu = \mu_0$ versus $H_1 : \mu > \mu_0$ with variance σ^2, the shaded area in the figure represents: (circle the best answer)

   (a) The probability of a type I error.

   (b) The probability of a Type II error if $\mu = \mu'$

   (c) The critical region.

   (d) The p-value.

   Solution: (b)

4. Let $\bar{X}$ be the mean of a random sample of $n = 36$ currents (in milliamperes) in a strip of wire in which each measurement has a mean of 15 and a variance of 9. Which of the following is the approximate value of $P(14.5 < \bar{X} < 15.5)$.

   (a) 0.8413

   (b) 0.8664

   (c) 0.3174

   (d) 0.6826

   Solution: (d)
   $\bar{X}$ has an approximate $N(15, 9/36)$ distribution.

$$P(14.5 < \bar{X} < 15.5) = P(\frac{14.5 - 15}{0.5} < \frac{\bar{X} - 15}{0.5} < \frac{15.5 - 15}{0.5})$$
$$\approx \Phi(1) - \Phi(-1) = 0.8413 - (1 - 0.8413) = 0.6826$$

5. Let $X_1, \ldots, X_5$ be independent Uniform(0,θ) distributed random variables, and let $Y = \max\{X_1, \ldots, X_n\}$ (i.e., Y is the n-th order statistic). To test $H_0 : \theta = 100$ against $H_0 : \theta > 100$, suppose we use a critical region $\{y > 98\}$. Find the probability of a Type-II error if $\theta = 150$.

(a) 0.67

(b) 0.12

(c) 0.05

(d) 0.23

Solution: (b)

$$P(Y < 98; \theta = 150) = P(X_1, \ldots, X_n < 98; X_1, \ldots, X_n \sim U(0, 150))$$
$$= \left(\frac{98}{150}\right)^5 = 0.12$$

6. Let $X^{(1)} < X^{(2)} < X^{(3)} < X^{(4)} < X^{(5)}$ be the order statistics associated with n independent observations X_1, X_2, X_3, X_4, X_5, each from the distribution with pdf $f(x) = 2x, 0 < x < 1$. What is the pdf of $X^{(2)}$?

   (a) $g_{(2)}(x) = 10x(1 - x^2)^3, \quad 0 < x < 1$

   (b) $g_{(2)}(x) = 40x^3(1 - x^2)^3, \quad 0 < x < 1$

   (c) $g_{(2)}(x) = 60x^5(1 - x^2)^3, \quad 0 < x < 1$

   (d) $g_{(2)}(x) = 50x^2(1 - x^2)^3, \quad 0 < x < 1$

   Solution: (b)

$$g_r(y) = \frac{n!}{(r-1)!(n-r)!}[F(y)]^{r-1}[1 - F(y)]^{n-r} f(y)$$

7. In a homework for STA2002, Xue Xi used a simple linear regression model to find a 95% confidence interval for $\mu(x)$ and a prediction interval for Y, both at $x = 68$. Not working very carefully, Xue Xi forgot which of the two resulting intervals is which. The two intervals computed are $[75.1, 85.1]$ and $[68.2, 88.2]$. Choose the correct answer:

   (a) $[75.1, 85.1]$ is the 95% confidence interval and $[68.2, 88.2]$ the 95% prediction interval.

   (b) $[68.2, 88.2]$ is the 95% confidence interval and $[75.1, 85.1]$ the 95% prediction interval.

   (c) At least one of these two intervals is incorrect.

   Solution: (c)
   The two intervals are not centered at the same value; the first is centered at 80.1 and the second at 78.2.

2 Long Questions

(72 points)

1. (6 points) Let $Y \sim \text{bin}(n, p)$. Then, the estimator $\hat{P} = Y/n$ is an unbiased estimator for the true population proportion p. Determine whether $n\hat{P}(1-\hat{P})$ is an unbiased estimator for the variance $\sigma^2 = np(1-p)$.

Solution:

$$\begin{aligned} E[n\hat{P}(1-\hat{P})] &= nE[\hat{P}] - nE[\hat{P}^2] \\ &= nE[\frac{Y}{n}] - nE[\frac{Y^2}{n^2}] \\ &= E[Y] - \frac{1}{n}E[Y^2] \\ &= np - \frac{1}{n}(np(1-p) + (np)^2) \\ &= (n-1)p(1-p) \end{aligned}$$

Not unbiased.

2. (20 points) Let $X_1, X_2, \ldots, X_n$ be random variables with unknown parameters μ and θ, $\theta > 0$, so that the probability density function is

$$f(x) = \begin{cases} \frac{1}{\theta} e^{-\frac{(x-\mu)}{\theta}} & x \geq \mu \\ 0 & x < \mu \end{cases}$$

(a) (10 points) Find the method-of-moments estimators of μ and θ.

(b) (10 points) Find the maximum likelihood estimators of μ and θ. (You may assume $x_1 \leq x_2 \leq \cdots \leq x_n$)

Solution:

(a)

$$E(X) = \int_{\mu}^{+\infty} \frac{x}{\theta} e^{-\frac{x-\mu}{\theta}} dx$$

Let $y = x - \mu$,then

$$E(X) = \int_{0}^{+\infty} \frac{1}{\theta}(y+\mu) e^{-\frac{y}{\theta}} dy = \theta + \mu$$

$$\begin{aligned} E(X^2) &= \int_u^{+\infty} \frac{x^2}{\theta} e^{-\frac{x-\mu}{\theta}} dx \\ &= \int_0^{+\infty} \frac{1}{\theta}(y+\mu)^2 e^{-\frac{y}{\theta}} dy = 2\theta^2 + 2\mu\theta + \mu^2 \\ &= \theta^2 + (\theta+\mu)^2. \end{aligned}$$

$$D(X) = E(X^2) - (EX)^2 = \theta^2$$

$$\begin{cases} \theta + \mu = M_1 = \bar{X}, \\ \theta^2 = M_2 = \frac{1}{n}\sum_{i=1}^n X_i - \bar{X}^2, \end{cases}$$

Therefore, $\hat{\mu} = \frac{1}{n}\sum_{i=1}^n X_i - \sqrt{\frac{1}{n}\sum_{i=1}^n (X_i - \bar{X})^2}, \hat{\theta} = \sqrt{\frac{1}{n}\sum_{i=1}^n (X_i - \bar{X})^2}.$

(b)

$$L(x_1, x_2, \cdots, x_n, \theta, \mu) = \frac{1}{\theta^n} e^{-\frac{1}{\theta}\sum_{i=1}^n (x_i - \mu)}, x_i \geq \mu$$

$$\ln L(x_1, x_2, \cdots, x_n, \theta, \mu) = -n \ln \theta - \frac{1}{\theta}\sum_{i=1}^n (x_i - \mu)$$

$$\frac{\partial \ln L}{\partial \mu} = \frac{n}{\theta} \neq 0$$

Since we see the derivative of the likelihood function with respect to μ is not 0, the maximum must occur at a boundary point of the region where the likelihood function is non-zero. Looking at the likelihood function, we see it is increasing in μ, but that it is 0 if μ is larger than one of $x_1, x_2, \ldots, x_n$. Hence, $\hat{\mu} = \min\{x_1, \ldots, x_n\} = x^{(1)}$ maximizes L.

$$\frac{\partial \ln L}{\partial \theta} = -\frac{n}{\theta} + \frac{\sum_{i=1}^n (x_i - \mu)}{\theta^2} = 0$$

$$\hat{\theta} = \bar{x} - x^{(1)}.$$

3. (12 points) Let 0.50,1.25,0.80,2.00 are randomly independent observations of X, given that $Y = \ln(X)$ has a $N(\mu, 1)$ distribution.

   (a) (4 points) Find a 95% confidence interval for μ.

   (b) (8 points) Find a 95% confidence interval for $E(X)$.

Solution

(a) The equation of confidence interval for μ should be

$$\left(\bar{Y} - \frac{\sigma}{\sqrt{n}} z_{\frac{\alpha}{2}}, \bar{Y} + \frac{\sigma}{\sqrt{n}} z_{\frac{\alpha}{2}}\right)$$

Since

$$\begin{aligned} P(|Z| < z_{\frac{\alpha}{2}}) &= 1-\alpha = 0.95 \\ z_{\frac{\alpha}{2}} &= 1.96, \sigma = 1, n = 4, \\ \bar{y} &= \frac{1}{4}(\ln 0.5 + \ln 1.25 + \ln 0.8 + \ln 2) = 0 \end{aligned}$$

Therefore, the 95% confidence interval for μ is $(-0.98, 0.98)$

(b) The pdf of Y is

$$f(y) = \frac{1}{2\pi} e^{-\frac{(y-\mu)^2}{2}}.$$

since $Y = \ln X, X = e^Y$, then

$$\begin{aligned} b = E(X) = E(e^Y) &= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^y e^{-\frac{(y-\mu)^2}{2}} dy \\ &= e^{\mu+\frac{1}{2}} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}[y-(\mu+1)]^2} dy \\ &= e^{\mu+\frac{1}{2}}. \end{aligned}$$

Therefore,

$$E(X) = e^{\mu+\frac{1}{2}}.$$

To derive a 95% confidence interval for $e^{\mu+\frac{1}{2}}$, recall that

$$P(\bar{Y} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \bar{Y} - 1.96\frac{\sigma}{\sqrt{n}}) = 0.95.$$

Since $e^{x+1/2}$ is a strictly increasing function of x, this implies that

$$P(e^{\bar{Y}-1.96\frac{\sigma}{\sqrt{n}}+\frac{1}{2}} < e^{\mu+\frac{1}{2}} < e^{\bar{Y}-1.96\frac{\sigma}{\sqrt{n}}+\frac{1}{2}}) = 0.95.$$

So a 95% confidence interval for $E(X)$ is $(e^{\bar{Y}-1.96\frac{\sigma}{\sqrt{n}}+\frac{1}{2}}, e^{\bar{Y}-1.96\frac{\sigma}{\sqrt{n}}+\frac{1}{2}})$. Using the provided data values, the 95% confidence interval for $E(X)$ is $(e^{-0.98+\frac{1}{2}}, e^{0.98+\frac{1}{2}})$, that is, $(e^{-0.48}, e^{1.48})$.

4. (13 points) Let X and Y equal the forces required to pull stud No. 3 and stud No. 4 out of a window that has been manufactured for an automobile. Assume that the distributions of X and Y are $N(\mu_X, \sigma_X^2)$ and $N(\mu_Y, \sigma_Y^2)$, respectively.

   (a) (2 points) If $m = n = 10$ observations are selected randomly, define a test statistic and a critical region for testing $H_0 : \mu_X - \mu_Y = 0$ against a two-sided alternative hypothesis. Let $\alpha = 0.05$. Assume that the variances are equal.

(b) (2 points) Given $n = 10$ observations of X, namely,

$$111 \quad 120 \quad 139 \quad 136 \quad 138 \quad 149 \quad 143 \quad 145 \quad 111 \quad 123$$

and $m = 10$ observations of Y, namely,

$$152 \quad 155 \quad 133 \quad 134 \quad 119 \quad 155 \quad 142 \quad 146 \quad 157 \quad 149$$

calculate the value of the test statistic and state your conclusion clearly. ($\bar{x} = 131.5, \bar{y} = 144.2, s_x^2 = 198.3, s_y^2 = 150.4$)

(c) (4 points) Find a 95% confidence interval for $\frac{\sigma_X^2}{\sigma_Y^2}$.

(d) (2 points) Based on your answer in (c), were you correct in assuming equal variances in (a)?

(e) (3 points) Construct box plots on the same figure for these two sets of data. Do the box plots confirm your decision in part (b)?

Solution

(a) Since $\sigma_X^2 = \sigma_Y^2 = \sigma^2$(unknown),therefore the test statistic should be

$$T = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{9S_x^2 + 9S_y^2}{18}(\frac{1}{10} + \frac{1}{10})}}$$

The critical region is $|T| \geq t_{0.025}(18) = 2.101$ (or, equivalently, $T < -2.101, T > 2.101$).

(b) The value of the test statistic is -2.151. Since $|-2.151| > 2.101$, we reject H_0;

(c) Since

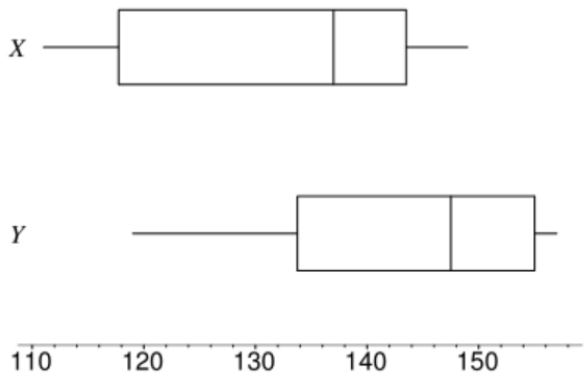
$$F = \frac{\sigma_Y^2}{\sigma_X^2} \cdot \frac{S_X^2}{S_Y^2} \sim F(n_1 - 1, n_2 - 1).$$

$$P\left\{F_{1-\frac{\alpha}{2}(n_1-1,n_2-1)} < \frac{\sigma_Y^2}{\sigma_X^2} \cdot \frac{S_X^2}{S_Y^2} < F_{\frac{\alpha}{2}}(n_1 - 1, n_2 - 1)\right\}$$

$$P\left\{\frac{S_X^2}{S_Y^2} \cdot \frac{1}{F_{\frac{\alpha}{2}}(n_1 - 1, n_2 - 1)} < \frac{\sigma_X^2}{\sigma_Y^2} < \frac{S_X^2}{S_Y^2} \cdot \frac{1}{F_{1-\frac{\alpha}{2}}(n_1 - 1, n_2 - 1)}\right\}$$

Therefore, the confidence interval for $\frac{\sigma_X^2}{\sigma_Y^2}$ should be (0.327,5.313)

(d) We can test $H_0 : \sigma_X^2 = \sigma_Y^2$ against $H_1 : \sigma_X^2 \neq \sigma_Y^2$ at significance level α by checking if the value $\frac{\sigma_X^2}{\sigma_Y^2} = 1$ is in the $(1 - \alpha)100\%$ confidence interval. Since it is contained in the 95% confidence interval, we would not reject H_0 at significance level $\alpha = 0.05$. So we used the correct test in (a) and (b).



(e)

$$\begin{array}{lll} \text{For} & X: & \widetilde{q}_1 = 117.75 \quad \widetilde{q}_3 = 143.5 \\ \text{For} & Y: & \widetilde{q}_1 = 133.75 \quad \widetilde{q}_3 = 155 \end{array}$$

5. (6 points) A headline read, "Majority of Chinese smokers have no plans to quit." The headline was based on a random sample of 100 Chinese smokers in which 57 stated they are not planning to quit. You are not convinced that the data really supports the conclusion that a majority (more than half) of Chinese smokers have no plans to quit, and you think the journalist should have done a hypothesis test. Perform this hypothesis test at significance level $\alpha = 0.05$. Make sure to give the p-value for your test, and to give the conclusion.

   Solution:

   Let p be the percentage of Chinese smokers who have no plans to quit. Then $H_0 : p = 0.5, H_1 : p > 0.5$. The test statistic is $z_0 = \frac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)/n}} = 1.40$. The p-value is $P(Z > 1.40) = 0.0808$. Since the p-value is greater than the significance level, we do not reject H_0. So the sample data does not give statistically significant evidence for the statement that a majority (more than half) of Chinese smokers have no plans to quit.

6. (15 points) Consider a linear regression model

$$Y_i = \alpha_1 + \beta x_i + \varepsilon_i, \quad i = 1, \ldots, n, \tag{1}$$

   where x_i are constants and $\varepsilon_i \sim N(0, \sigma^2)$ independently for all i. Suppose α_1, β and σ^2 are unknown.

   One can show that the MLE of α_1, β and σ^2 are:

$$\hat{\beta} = \frac{\sum_{i=1}^n (x_i - \bar{x})Y_i}{\sum_{j=1}^n (x_j - \bar{x})^2}, \quad \hat{\alpha}_1 = \bar{Y} - \hat{\beta}\bar{x}, \quad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n (Y_i - \bar{Y} - \hat{\beta}(x_i - \bar{x}))^2,$$

where $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$.

(a) (10 points) Find the distributions of $\hat{\alpha_1}$ and $\hat{\beta}$.

(b) (5 points) We want to test $H_0 : \alpha_1 = 0$ against $H_1 : \alpha_1 \neq 0$. Give the test statistic and state its distribution if H_0 is true. Then give the critical region for the test at significance level γ .
(You may use, without proof, the fact that $\frac{n\hat{\sigma}^2}{\sigma^2} \sim \chi^2(n-2)$ and that $\hat{\alpha_1}$ and $\hat{\sigma}^2$ are independent.)

Solution:

(a) $\hat{\alpha_1}$ and $\hat{\beta}$ are both linear function of independent and normally distributed random variables $Y_1, Y_2, \ldots, Y_n$ where $Y_i \sim N(\alpha_1 + \beta x_i, \sigma^2)$. Therefore, $\hat{\alpha_1}$ and $\hat{\beta}$ are both normally dsitributed. We can compute their means and variances

for $\hat{\beta}$

$$\begin{aligned} E(\hat{\beta}) &= \frac{\sum_{i=1}^{n}(x_i - \bar{x})E(Y_i)}{\sum_{j=1}^{n}(x_j - \bar{x})^2} \\ &= \frac{\sum_{i=1}^{n}(x_i - \bar{x})(\alpha_1 + \beta x_i)}{\sum_{j=1}^{n}(x_j - \bar{x})^2} \\ &= \frac{\alpha_1 \sum_{i=1}^{n}(x_i - \bar{x}) + \beta \sum_{i=1}^{n} x_i(x_i - \bar{x})}{\sum_{j=1}^{n}(x_j - \bar{x})^2} \\ &= \frac{\alpha_1 \sum_{i=1}^{n}(x_i - \bar{x}) + \beta \sum_{i=1}^{n}(x_i - \bar{x})(x_i - \bar{x})}{\sum_{j=1}^{n}(x_j - \bar{x})^2} = \beta \end{aligned}$$

$$\begin{aligned} Var(\hat{\beta}) &= \sum_{i=1}^{n}\left[\frac{x_i - \bar{x}}{\sum_{j=1}^{n}(x_j - \bar{x})^2}\right]^2 Var(Y_i) \\ &= \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{\left[\sum_{i=j}^{n}(x_j - \bar{x})^2\right]^2}\sigma^2 = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \end{aligned}$$

for $\hat{\alpha_1}$

$$\begin{aligned} E(\hat{\alpha_1}) &= E(\bar{Y} - \hat{\beta}\bar{x}) = E(\bar{Y}) - E(\hat{\beta}\bar{x}) \\ &= \frac{1}{n}\sum_{i=1}^{n}(\alpha_1 + \beta x_i) - \frac{1}{n}\sum_{1}^{n} x_i\beta = \alpha_1 \end{aligned}$$

$$Var(\hat{\alpha_1}) = Var(\bar{Y} - \hat{\beta}\bar{x}) = = Var(\frac{1}{n}\sum_{i=1}^{n} Y_i - \frac{\bar{x}\sum_{i=1}^{n}(x_i - \bar{x})Y_i}{\sum_{j=1}^{n}(x_j - \bar{x})^2})$$

$$= (\frac{1}{n})^2 \sum_{i=1}^{n} Var(Y_i) + \sum_{i=1}^{n} \left[\frac{x_i - \bar{x}}{\sum_{j=1}^{n}(x_j - \bar{x})^2}\right]^2 \bar{x}^2 Var(Y_i)$$

$$= \frac{\sigma^2}{n} + \frac{\sigma^2 \bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2})$$

Therefore we get

$$\hat{\alpha_1} \sim N(\alpha_1, \sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2})), \quad \hat{\beta} \sim N(\beta, \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}).$$

(b) Since we have

$$\frac{\hat{\alpha}_1 - \alpha_1}{\sigma\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}} \sim N(0,1), \quad \frac{n\hat{\sigma}^2}{\sigma^2} \sim \chi^2(n-2),$$

and they are independent. Therefore, under $H_0 : \alpha_1 = 0$

$$T = \frac{\hat{\alpha}_1}{\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}\sqrt{\frac{n\hat{\sigma}^2}{n-2}}} \sim t(n-2).$$

The critical region is of the form $|T| > t_{\gamma/2}(n-2)$ for $H_1 : \alpha_1 \neq 0$.